



UNIVERSITY OF
BIRMINGHAM

Superconductivity

Lecture 17 – “Lattice, Pairing and Gap”

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Overview

A hand-waving description

... of what's happening with a superconductor is like ...

In the remaining weeks

- Experimental clues
- Many-body Hamiltonian
- Ground states & excitations
- Experimental confirmation

In more detail ...

- Experimental clues
 - Lattice - phonons involved
 - Pairing - two electrons involved
 - Gap - the ground state protected
- Many-body Hamiltonian
 - $\mathcal{H} = \sum_{\mathbf{k}, \sigma} \epsilon_{\mathbf{k}} n_{\mathbf{k}\sigma} - |V| \sum_{\mathbf{k}, \mathbf{k}'} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} c_{-\mathbf{k}'\downarrow} c_{\mathbf{k}'\uparrow}$
- Ground states & excitations
 - $|BCS\rangle = \prod_{\mathbf{k}} \left(u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^{\dagger} c_{-\mathbf{k}\downarrow}^{\dagger} \right) |0\rangle.$
 - $\mathcal{H}_{\text{MF}} = \sum_{\mathbf{k}} E_{\mathbf{k}} \left(\gamma_{\mathbf{k}\uparrow}^{\dagger} \gamma_{\mathbf{k}\uparrow} + \gamma_{-\mathbf{k}\downarrow}^{\dagger} \gamma_{-\mathbf{k}\downarrow} \right)$
- Experimental confirmation
 - Specific heat, tunneling, ultrasound attenuation, NMR, etc.

Lattice

Isotope effect

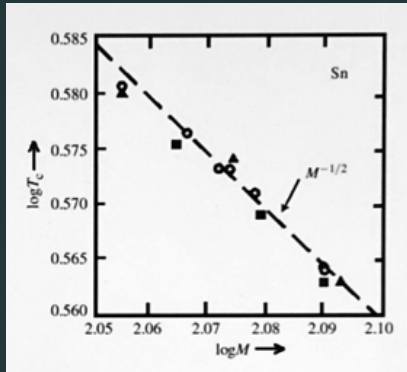


Figure 1: T_c of Sn as a function of isotope enrichment (M , the average mass of Sn ions).

$$T_c \propto M^{-1/2} \Leftrightarrow \text{Debye temp. } \theta_D \propto M^{-1/2} \quad (1)$$

Pairing

Flux quantum

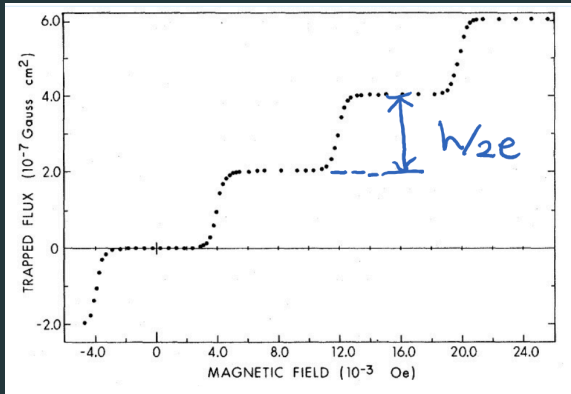


Figure 2: Trapped flux quanta as a function of an applied magnetic field.

Pairing via lattice

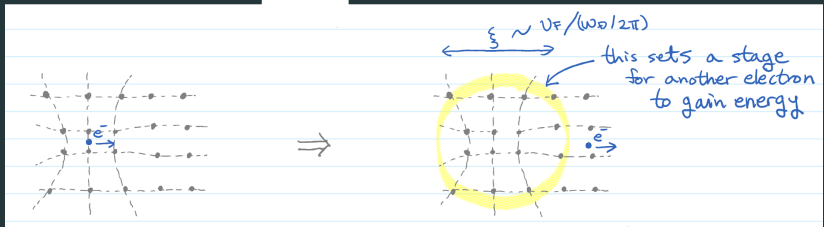


Figure 3: A possible scenario for pairing two electrons via ionic lattice vibration.

Gap

A horizontal line spanning the width of the slide, positioned below the word 'Gap'. The line is composed of two segments: an orange segment on the left and a grey segment on the right.

Recall heat capacity

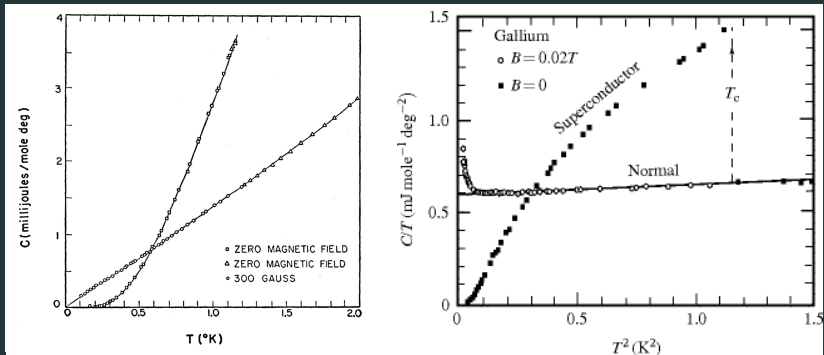


Figure 4: Left: Aluminum; Right: Gallium

C_e suppressed more than T^3 at low $T \Rightarrow$ exponentially...?

Specific heat

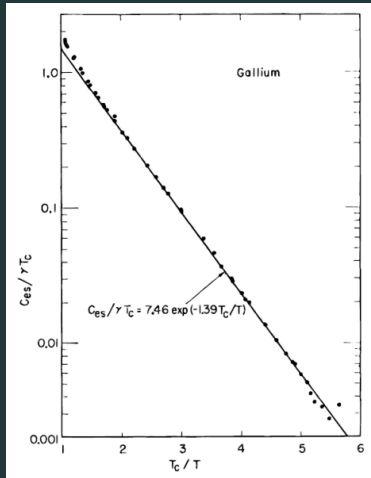


Figure 5: Electronic specific heat of Gallium in an Arrhenius plot

- Straight line in Arrhenius plot
 $\Rightarrow$ a gapped behaviour
 $\Leftrightarrow C_e \propto \exp(-\Delta/T)$
- Why T_c/T and $C_e/\gamma T$ for x and y axes, respectively?
(rather than $1/T$ and C_e)
- Some deviation close to T_c .
 $\Rightarrow T$ evolution of Δ

T evolution of gap

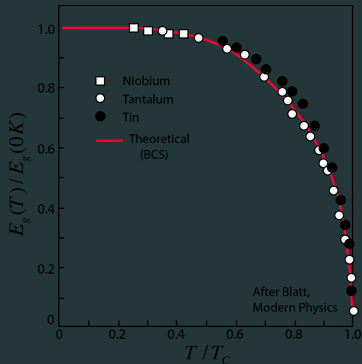
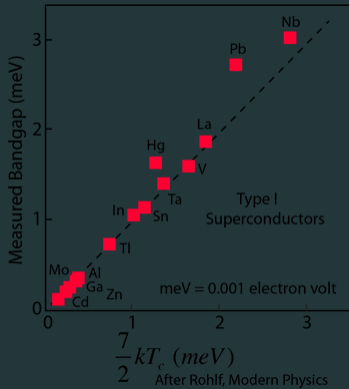


Figure 6: Temperature dependence of gap size for Mo, Ta and Sn.

- Why T/T_c and $E_g(T)/E(0\text{ K})$ for x and y axes, respectively?
- All SCs would have different T_c and $E_g (= 2\Delta)$.
- There must be a universal relation between T_c and Δ .

$\Delta(0)$ and T_c



- Universally, $k_B T_c \simeq 1.75 \Delta(0)$.

Lattice, Pairing and Gap

A simple minded picture:

- Pairing via lattice;
- Gap to breaking pairs (energy cost).